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Product functions:

It meets the functional requirements in TSG T7001-2023 Regulation for Lift Supervisory Inspection and Periodical Inspection, and is designed based on the daily work content of inspectors to better assist inspectors in their work, thus greatly reducing the labor intensity of inspectors and improving work efficiency. Meanwhile, the data connection with user-owned platform can be realized via the expandable remote interaction and background expert guidance mode, connection with special equipment management system and the open communication protocol.

Performance characteristics:

- AR glasses can perform the native functions of screen sharing, audio-video interaction, AR tagging, multi-terminal collaboration, etc. In addition, the operating system is provided with relevant materials required by inspectors in their daily work, including laws, regulations, administrative rules and regulations, and technical specifications and standards on safety. When working, AR glasses are used for inspector identification and identification of elevator uniqueness sign. The audios and videos in the whole inspection process will be displayed on the main screen of AR glasses in real time and include images of a full-screen main camera and images of three small-screen auxiliary cameras and images of the small-screen auxiliary cameras correspond to the machine room, car and pit area respectively.
- Auxiliary camera 1: It is installed in the machine room area, equipped with magnetic base and its own power supply. With voice call function, users can make real-time calls with AR glasses. The camera is mainly used during the detection such as audio-visual recording of main drive unit detection (starting, running and stopping), load test, governor detection.
- Auxiliary camera 2: It is installed in the car area (inner, outer roof, outer bottom, etc.). This camera is mainly used for audio-visual recording of the car-related detection. Moreover, the auxiliary camera 2 can optionally be equipped with an elevator running condition detector, and the elevator speed, running height, acceleration and other related information can be transmitted to the operating system of AR glasses in real time.
- Auxiliary camera 3: It is installed in the pit area, equipped with magnetic base and its own power supply. With voice call function, users can make real-time calls with AR glasses. It can perform lighting function. The camera is mainly used for the pit detection via audio-visual recording, such as buffer compression and resetting.
- Other external equipment: Other testing equipment produced by Dalian Easy Test Technology Co., Ltd. can be connected to the control system of AR glasses, such as balance coefficient detector and other equipment required for elevator carriage supervision and inspection and regular inspection of carriage quality and reset. All the detection data can be stored in a database by one-time operation with one control terminal, which greatly facilitates the operation of the detection personnel, and the detection data is not recorded repeatedly, thus improving the detection efficiency and reducing the labor intensity.

电梯能耗测试仪 ET-NH
Elevator energy consumption tester ET-NH

Product functions:

It applies to the measuring of energy consumption, energy efficiency and standby energy consumption of elevators and escalators. The built-in high-precision voltage and current detection module can quickly complete the analysis of running energy consumption or standby timing energy consumption. It can be used to measure parameters such as energy consumption under different loading conditions, evaluation index of elevator running energy efficiency, standard standby energy consumption and standby energy consumption with automatic sleep technology. The measurement is accurate, efficient and reliable, and the device is a necessary new detecting instrument for special inspection and elevator maintenance units.

Performance characteristics:

- The main measuring device is configured with high-precision voltage and current sensors and communicates with the handheld terminal through USB.
- The main measuring device is equipped with a power indicator to display the power in real time;
- It is equipped with proprietary data analysis and processing software, and this is intuitive and convenient, ensuring easy analysis and operation.
- The host shell is made of light aluminum alloy, which is light and solid;
- The whole measurement process is completed in the machine room, and the results can be observed in real time, hence safe and efficient detection;
- The detector is provided with file management function, and users can save, view, print and delete data conveniently;
- Mass data storage function;
- Chinese characters, numbers and special characters can be input at will, and the setting can be completed conveniently and flexibly;
- Automatic triggering is supported, and triggering operation will not cause related errors.

Technical parameters:

Range of energy consumption:	1-9999M,precision:±0.5%
Frequency range of power measurement:	DC, 0.5HZ~500HZ
Measuring voltage of variable frequency power sensor:	0~500VAC
Measuring current of variable frequency power sensor:	0~50A
Measuring precision of variable frequency power sensor:	≥0.5%
Measuring current of clamp-type current sensor:	AC50A
Measuring precision of clamp-type current sensor:	≥0.5%
Lifting height of elevator:	0~200m, ≤1%

电梯钢丝绳滑移量检测仪 ET-HYC
Elevator wire rope slip detector ET-HYC

Product functions:

This detector is the special equipment for detecting the elevator wire rope slip, and is used to detect the slip of elevator traction wire rope on the traction sheave in case of sudden emergency stop during elevator operation, and the maximum operating distance and speed of wire rope and traction sheave during elevator operation.

Performance characteristics:

- The sensors used in each module of this instrument are high-precision, and feature high sampling precision and fast sampling speed, effectively improving the measuring precision of the instrument;
- The instrument does not intervene in the elevator control system during the measurement and does not affect the normal operation of the elevator;
- Two speed-time 2D curves of traction sheave and wire rope are generated in real time, and the measurement process is more intuitive and accurate;
- Control system platform: Windows operating system. It can intelligently analyze and generate the measurement report after the completion of measurement, the measurement data has its own time attribute, and the system can print the test report on site by wireless printer;
- USB2.0 communication, no driver installation, and easy data sharing;
- External 4-segment battery power display indicator;
- The wireless transmission mode is adopted for the handheld terminal and the main measuring device, and no additional signal transmission antenna is required;
- The host shell is made of light aluminum alloy, which is light and solid;
- The detector is provided with file management function, and users can save, view, print and delete data conveniently.

Technical parameters:

Speed measuring range:	0~10m/s (0~6000r/min)
Speed measuring precision:	≤1%
Measuring precision of slip:	≤±0.1mm.
Speed display resolution:	0.001m/s
Distance display resolution:	0.001m
Number of saved test groups:	No less than 99,999 groups

电梯限速器测试仪 ET-X1

Elevator governor tester ET-X1

Product functions:

This instrument is used to detect the electrical speed and mechanical speed of elevator governor. It applies to the on-site inspections of elevator governor by special equipment inspection and testing organizations, elevator companies and elevator maintenance units.

Performance characteristics:

- It applies to the inspection status of elevator governor in China and conforms to the use habits of on-site inspectors. It meets the requirements of relevant national inspection regulations and can comprehensively record the key data and information of elevator governor inspection;
- Touch screen display interface, which supports the input of Chinese characters, numbers and characters at will, features simple interface and easy operation, and displays the remaining power;
- Two measurement modes: contact and non-contact measurement (Hall sensor speed measurement), are for choice;
- The non-contact measurement is realized by using a Hall sensor, which is adsorbed on the side plate of the governor by a magnetic suction device, and the magnetic bean is adsorbed on the governor wheel, hence convenient and fast installation;
- Operation, display, driving and speed measurement devices are designed to be handheld and integrated, and it is unnecessary to connect a handheld measuring terminal on site;
- Built-in 24V lithium battery, indicating no need for external power supply on site, hence safe operation and stable control;
- The uplink and downlink measurement data can be recorded three times according to the requirements of inspectors and inspection regulations;
- The host shell is made of light aluminum alloy, which is light and solid;

Technical parameters:

Speed measuring precision:	±0.01m/s	Touch-screen display:	3.5-inch touchscreen
Measuring range:	0~3.5m/s	Battery capacity:	3200mAH
Speed measurement mode:	Contact/non-contact type	Size:	240mm*78mm*71.5mm

电梯限速器校验台 ET-XT
Elevator Governor Calibration Console ET-XT

Product functions:

It applies to off-site inspection aiming to measure the electrical action speed, mechanical action speed, lifting force and other parameters of various elevator governors. It also applies to the calibration and inspection of elevator governor manufacturers and related users, a test report can be generated after the test is completed, and the external printer is provided. This device is a comprehensive calibration instrument.

Performance characteristics:

- The instrument is provided with touch screen display interface, which supports the input of Chinese characters, numbers and characters at will, and features simple interface and easy operation;
- Various stop modes can meet the measurement requirements of different types of governors;
- Range: The driving speed can reach up to 10 m/s;
- Control system platform: windows operating system, which can intelligently analyze and generate measurement reports after the measurement is completed;
- Software requirements: integrated design, data analysis, report printing, data storage, etc.
- Data transmission mode: USB communication;
- Power supply mode: 220V AC input;
- The screen is dust-proof, waterproof and scratch-proof, and features long service life, high accuracy, light transmittance > 98% and fast response.
- It is equipped with a special support frame, the driving position can be adjusted in three dimensions, and the acceleration can be adjusted automatically according to the rated speed so that the detection can be completed quickly.

Technical parameters:

Speed measuring range:	≤10m/s
Measurement error:	<0.01m/s
Speed regulation range:	0.2m/s~10m/s
Speed display resolution:	0.001m/s
Measuring range of lifting force:	≤500kg
Maximum speed of driving motor:	3000r/m
Lifting force display resolution:	0.1N

Product functions:

It is designed to measure parameters such as triaxial acceleration, jerk, A95 acceleration, speed, braking distance and displacement during the operation of machine-room elevator and machine-room-less elevator.

Performance characteristics:

- The main measuring device is a 3D-MEMS acceleration sensor, and the cross-axis sensitivity and lateral vibration sensitivity are 1%.
- The main measuring device is equipped with a 0.87-inch small LCD screen, which displays the three-axis acceleration and battery power in real time;
- The data analysis and processing software provided can realize convenient multi-point measurement and data analysis, and displays triaxial vibration curve, triaxial acceleration and running speed in real time in the measurement process, and this is intuitive and convenient, ensuring easy analysis and operation.
- The analysis diagram of vibration curves of three axes - X, Y and Z axles displayed in real time during the measurement can be zoomed in and out, which is convenient for observing the operation state at site.
- The whole measurement process is completed in the car, and the numerical changes at different sampling points can be analyzed through the curve chart, and the vibration can be automatically monitored, with a resonance frequency of 2.4 KHz and a bandwidth of (3DB) 500 Hz;
- It is equipped with system setting function aiming to check and view the calibration detection and curve displacement variation detection before detection, and is configured with the port setting function and configuration information. A three-axis vibration curve area is designed for the calibration detection, so as to facilitate the check of the field use, link and signal interference of the instruments.
- The impact resistance of the sensor is 100g@11ms, the same for three axes (half sine wave);
- Output rate of triaxial vibration sensor: 5 Hz, 10 Hz, 25 Hz, 50 Hz, 100 Hz, 200 Hz, 500 Hz and 1,000 Hz.

Technical parameters:

Number of axles of sensor:	Three axis directions (X, Y, Z)
Measuring precision of acceleration:	±1mg
Measuring range of acceleration:	±2g
Output signal:	TTL
Printing method:	Wireless Bluetooth printing
Impact resistance:	2000G
Deviation temperature coefficient:	0.01%/°C

电梯振动分析仪 ET-J4
Elevator vibration analyzer ET-J4

Product functions:

It is designed to measure parameters such as triaxial acceleration, jerk, A95 acceleration, speed, braking distance and displacement during the operation of machine-room elevator and machine-room-less elevator.

Performance characteristics:

- The instrument is of integrated design without any externally connected wires, tablets or operators, compact structure and convenient adjustment;
- The main device is equipped with a 4.3-inch LCD, which displays the three-axis vibration acceleration curve in real time;
- The special data analysis and processing software provided can realize convenient multi-point measurement and data analysis, and displays triaxial vibration curve, triaxial acceleration and running speed in real time in the measurement process, and this is intuitive and convenient, ensuring easy analysis and operation;
- The analysis diagram of vibration curves of three axes - X, Y and Z axles displayed in real time during the measurement can be zoomed in and out, which is convenient for observing the operation state at site.
- The whole measurement process is completed in the car, and the numerical changes at different sampling points can be analyzed through the curve chart, and the vibration can be automatically monitored;
- The instrument is equipped with system setting function aiming to check and view the calibration detection and curve displacement variation detection before detection, and is configured with a display indicating the maximum and minimum values of triaxial acceleration. A three-axis vibration curve area is designed for the calibration detection, so as to facilitate the check of the field use, link and signal interference of the instruments.

Technical parameters:

Screen size:	4.3 in	Measuring precision of acceleration:	0.5%F.S
Measuring range of acceleration:	±2g	Non-linearity:	0.1%F.S
Number of axles:	Three axes (X, Y, Z)	Measuring range of frequency:	5Hz-100Hz

电梯钢丝绳张紧力测试仪 ET-GZ
Elevator wire rope tension tester ET-GZ

Product functions:

It is designed to measure the tension and balance coefficient of wire ropes of various elevators, and can also be used to adjust the tension uniformity among multiple wire ropes.

Performance characteristics:

- The main measuring device is a high-precision pressure sensor, and features the integrated designed of measurement, display and operation. There is no need to connect a handheld measuring terminal or a tablet computer on site, thus more convenient and faster operation.
- The tension of 10 wire ropes can be measured one by one;
- Static measurement of no-load elevator balance coefficient is safe, efficient and accurate.
- It can monitor and display the battery power state;
- The LCD backlight is turned off automatically after no operation for more than 30 s, and the device is turned off automatically after no operation for more than 15 min. The device features low power consumption design and can run for 10 h on a single charge.
- The main engine shell is made of light aluminum alloy, which is light and strong; the detection sensor features accurate data acquisition;

Technical parameters:

Force measuring range:	0~500N	Repeatability error:	≤0.5%F.S
Force measuring precision:	≤1%F.S	Safety overload:	120%F.S
Force measurement resolution:	0.1N	Balance coefficient measuring precision:	≤2%
Nonlinear	≤0.5%F.S	Range of wire rope diameter:	4mm~16mm

电梯 125%载荷制动试验保护装置 ET-ZH
Elevator 125% load braking test protection device ET-ZH

Product functions:

It applies to 125% load braking test of the elevator. When the elevator brake cannot complete braking and the car continues to slide down, the equipment can protect the car and the elevator effectively; meanwhile, it can detect the braking distance, braking deceleration, vibration peak and other parameters of the elevator.

Performance characteristics:

- The main device is equipped with high-precision speed and displacement sensors;
- high-power spread-spectrum wireless communication is conducted between the main device and terminal, inductive switch of safety loop and push-pull electromagnetic induction protection switch, and the transmission distance can reach 200 m based on the premise that the wall and car wall are penetrated;
- The device is equipped with special data analysis and processing software, which supports data filtering, curve display and analysis, and it is intuitive and convenient, ensuring easy analysis and operation.
- The whole measurement process is completed in the machine room or on the top floor, ensuring safe and efficient detection;
- It can display the operating distance, braking slip and operation curve in real-time, and conduct real-time analysis;
- The push-pull electromagnetic induction protection switch is adopted to prevent the elevator car from slipping or falling;
- Handheld terminal is a high stability system rather than Windows, IOS, Android system, or mobile phone or PAC;
- In order to ensure the communication quality, the terminal shell is made of high-strength nonmetallic material;

Technical parameters:

Response time of push-pull electromagnetic induction protection switch:	5ms	Update frequency:	10 times/s
Response time of inductive switch of safety loop:	5ms	Wireless communication mode:	Spread-spectrum wireless transmission
Measuring precision of acceleration:	±0.01m/s2	Wireless communication distance (the wall and car wall can be penetrated):	200 m
Displacement measuring range:	200 m	Operating temperature:	-30°C-50°C

电梯门锁啮合深度检测仪 ET-MN
Elevator door lock engagement depth detector ET-MN

Product functions:

It is applicable for detecting the engagement depth of elevator door lock. The design of the instrument makes full use of ergonomics, is based on high-precision encoder and on-off detection, and records the engagement depth of locking elements at the closing and opening moment of elevator door lock, which conforms to the use habits of personnel, has high compatibility and strong stability, and is more suitable for industrial field use.

Performance characteristics:

- Operation, display, driving and speed measurement devices are designed to be handheld and integrated, and it is unnecessary to connect a handheld measuring terminal on site;
- The instrument is of integrated design, without any externally connected wires and external equipment, compact structure, convenient adjustment, and is convenient for operation in narrow space;
- The instrument is of high precision, and designed according to the working condition of the arc rising or falling of the door lock opening and braking. The measurement can be completed by pressing only one key, the vertical displacement can be calculated without calculating the angle and producing systematic errors;
- The instrument can simulate a WiFi hotspot and link with the computer via LAN, and the stored data can be directly transmitted to the computer. Open the file browser and enter the IP address displayed on the instrument to directly find the stored file data and print it.
- The instrument can display the measured data in real time, and automatically record the data and the engagement depth of the door lock when the trigger switch is turned off;
- The instrument shall avoid the limitation in service environment, and shall be in the physical key mode instead of the touch key mode;
- The mechanical detection arm of the instrument can be divided into fixed and mobile modes. In order to match with the detection form, the fixed claw is divided into 4 sets, and the mobile claw is divided into 5 sets;
- The mobile claw and the fixed claw can be matched in pairs for different types of door locks;
- Control system platform: Rtos operating system, real-time display on 1.5-inch screen, dynamic display of displacement via 2 physical buttons and the displacement measurement button, and value recording;
- The wireless printer prints the test report on site;
- Low-power processor is adopted to minimize power consumption, with reverse connection and load short-circuit protection and electrical detection line;

Technical parameters:

Displacement measuring range:	0~20mm
Measuring precision:	0.07mm
Measurement resolution:	0.01mm
Maximum response frequency:	Conductor leading-out type

电梯层门导靴啮合深度动态检测仪 ET-MD
Dynamic detector for engagement depth of elevator landing door guide shoes ET-MD

Product functions:

It applies to the on-line dynamic detection of engagement depth of elevator landing door guide shoes, and can display measured values and curves in real time. The information of engagement depth exceeding the standard line can be seen visually. The instrument features convenient installation, quick operation and accurate detection.

Performance characteristics:

- The integrated algorithm of high-precision CMOS image sensor and displacement sensor is adopted so that the distance setting reflective sensor realizes high-precision measurement and high stability.
- The instrument and tooling fixture are of integrated design, and the stroke of fixture can be adjusted within the range of 8 mm - 43 mm;
- In order to ensure convenient installation and applicability of the instrument, the effective free stroke (the maximum stroke is 35 mm) is designed to be locked at any position within the effective stroke.
- The datum plane and auxiliary datum plane are in two-section connection, with size of datum plane and auxiliary datum plane being 120 mm and 150 mm respectively;
- The engagement depth and displacement are measured synchronously, and the displacement detector is contacted in the elastic contact mode to ensure the continuity and accuracy of data reception; the maximum longitudinal travel range is about 48 mm within the elastic range;
- It is configured with system setting function so as to mark calibration and detection before detection, view the detection and verification of the curve displacement change in real time, and configure port setting and configuration information;
- The initial function is set before the measurement starts, and this facilitates the inspection of on-site use, link and signal interference of the instrument.
- During the detection, both the left and right height operation curves can be displayed, and the displacement equivalent can be observed and recorded at site, and this facilitates the on-site re-inspection, view of detection position and data comparison after measurement.

Technical parameters:

Measuring range:	0~70mm
Measuring precision:	≤0.07mm
Temperature characteristics:	±0.03%F·S/°C
Short circuit protection:	Automatic recovery type

电梯层门导靴啮合深度动态检测仪 ET-MD
Detection Ruler for Engagement Depth of Elevator Landing Door Guide Shoes ET-MDC

Product functions:

It applies to the static detection of engagement depth of elevator landing door guide shoes, and can display measured values. The information of engagement depth can be seen visually via the actual readings. The instrument features convenient installation, quick operation and accurate detection.

Performance characteristics:

- The product is of integrated design, can complete the independent detection process without configuring the terminal and antenna;
- The datum plane and auxiliary datum plane of the instrument are in three-section connection, with size of datum plane and auxiliary datum plane being 115 mm and 200 mm respectively. The auxiliary plane reduces the space occupation, and the instrument features convenient installation and easy operation, and is suitable for most personnel.
- The instrument is fixed with handrails, and can be installed conveniently and is suitable for most elevator landing doors in the market;
- It can be detected directly without calibration, and the detection result is the result of engagement depth;
- The detection position can be measured at any selected position of the landing door;
- It is configured with system setting function so as to mark calibration and detection before detection, view the detection and verification of the curve displacement change in real time, and configure port setting and configuration information. The initial function is set before the measurement starts, and this facilitates the inspection of on-site use, link and signal interference of the instrument;
- During the detection, both the left and right height operation curves can be displayed, and the displacement equivalent can be observed and recorded at site, and this facilitates the on-site re-inspection, view of detection position and data comparison after measurement.

Technical parameters:

Measuring range:	0~50mm
Measuring precision:	≤0.1mm
Resolution:	0.01mm

电梯轿厢与井道壁距离测量仪 ET-JD
Measuring instrument of distance between elevator car and shaft wall ET-JD

Product functions:

It is used to measure the distance between the elevator car and the shaft wall and detect the out-of-tolerance, so as to prevent unsafe factors caused by the long-term operation of the elevator.

Performance characteristics:

- Using high-precision laser radar scanning scheme, 360° omnidirectional laser ranging scanning is conducted within the 12 m radius in 2D plane, and the planar point cloud map information of the space is generated;
- The mechanical part of the mechanical ranging core rotating at high speed starts to rotate clockwise for scanning after the power is supplied for each system;
- Special software with displacement programming can be used to adjust the displacement deviation of the instrument, and it is suitable for most elevators in the market.
- As the instrument has its own speed detection and adaptive system, the scanning frequency will be adjusted automatically based on the actual operation;
- Adopting the laser triangulation ranging technology and high-speed visual acquisition and processing mechanism, the instrument completes ranging actions of no less than 8,000 times per second; real-time calculation is performed by the receiving processor of visual acquisition system;
- An operation sampling diagram can be formed during the detection, and the visual range of the sampling diagram can be adjusted within the range of 100 mm - 1,600 mm, and standard lines are for reference;
- Output data: the sampling measurement value is the current actual distance measurement, the angle is calculated as the scanning angle of the current orientation distance of the adjustable included angle, and the data is the starting and ending positions of the 180° included angle boundary for effective detection;
- As the instrument has built-in laser power consumption detection and sensor health check logic, it can automatically stop scanning and turn off the laser transmitter in case of failure, so as to avoid damage to the outside world and itself;
- When the sampling period is 1,450 points/week, the scanning frequency reaches 5.5Hz, and the highest scanning frequency can reach 10Hz.
- Low-power (< 5 mW) infrared laser serves as the emission light source, and is driven in adjustable pulse mode, and the laser only emits within a very short time;
- The safety performance of radar sensor for laser scanning can reach the laser safety standard of Class I; the product complies with 21 CFR 1040.10 and 1040.11;
- When measuring, the instrument must be placed at the top of the car without fixed installation; the position can be adjusted at will for measuring; the measuring methods include surface-based detection or point-based detection;

Technical parameters:

Ranging distance:	0.15m~12m	Angle resolution:	≤ 1°
Ranging resolution:	<0.5mm	Displacement measuring range:	≤ 200m
Measuring precision:	±1mm	Displacement measuring precision:	±2.5mm
Range of scanning angle:	0~360°	Data Sampling frequency:	≥ 8000Hz
Range of adjustable angle:	0~180°		

Elevator shaft wall strength detector ET-JNT

电梯井道壁强度检测仪 ET-JNT
Elevator shaft wall strength detector ET-JNT

Product functions:

This instrument can be used by inspection and testing departments of special equipment for the strength detection of elevator shaft wall. The equipment is developed according to GB T 7588.1-2020 Safety Rules for the Construction and Installation of Lifts - Part 1: Passenger and Goods Passenger Lifts.

Performance characteristics:

- The instrument can adapt to different well widths, applies vertical pressure to the shaft wall by extrusion, and measures the deformation of the shaft wall at the same time, so as to judge whether the deformation and pressure value meet the requirements of relevant specifications;
- It is configured with double alarm functions of upper limits of force application and deformation.
- The instrument has an internal lithium battery for power supply, and no external power supply is required;
- Operation of touch screen;
- The detection results are stored in the internal memory for easy viewing, deletion and other operations;
- The instrument is equipped with a Bluetooth printer, and the detection results can be printed on site.
- The outer packaging is ABS shock-proof toolbox, and can be is taken out conveniently.

Technical parameters:

Measuring precision of deformation:	±0.1%F.S	Measuring range of force:	0-2000N
Measuring range of deformation:	0-20mm	Measuring precision of force:	≤ 1.0%F
Adjustable length of support rod		Display screen:	Touch screen
Mode of force application:	Electric loading	Printing method:	Bluetooth printer

电梯平衡系数检测仪 ET-DP
Elevator balance coefficient detector ET-DP

Product functions:

This instrument can perform on-line on-site detection of various elevator balance coefficients under no load condition; it can detect the machine-room-less elevator and steel belt elevator, and can detect all kinds of elevators in VVVF and worm and gear control mode without selecting the control mode.

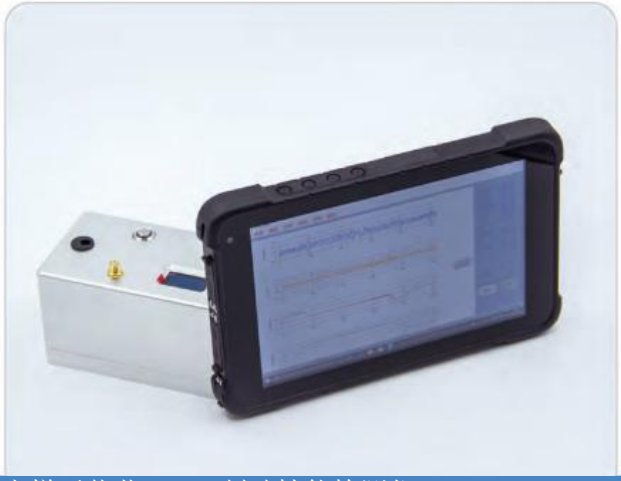
Performance characteristics:

- The balance coefficient detection can be completed by running the elevator up and down once each (in no particular order), the balance coefficient is calculated by default according to the last two up and down data, and the last data is automatically replaced;
- The host is equipped with an open-close anti-misjudgment device or adopts an anti-misjudgment switch;
- It is equipped with system setting function. The printer connection and printing test are set before testing, and port setting and configuration information are configured, which is convenient for checking on-site use, link and signal interference of instruments.
- It can be used to detect elevator with permanent magnet gearless tractor, elevator with worm and gear tractor and elevator without machine room, etc.
- The measuring interface can display the up and down power change curves of the tractor in real time;
- The measured data shows the voltage change value, current change value, operating power value and total power value of the elevator in real time;
- The measurement results include: 1) the up and down midpoint power and balance coefficient of the elevator with permanent magnet gearless tractor; 2) the no-load and on-load down midpoint power and balance coefficient of the elevator with worm and gear tractor;
- The instrument is designed according to IEC61010-1 safety standard, and the safety specification refers to Saftey Class I;
- The instrument uses 32-bit ARM microprocessor, adopts high-speed and high-precision A/D converter as the core instrument, has the function of high-precision, high-stability and multiple-parameter measurement, and is equipped with USB communication;
- The instrument consists of anti-aliasing low-pass filtering circuits with automatic signal amplification & acquisition and automatic measuring range processing;
- The speed measuring module is independently configured with 3D-MEMS acceleration sensor, and the cross-axis sensitivity and lateral vibration sensitivity are 1%.
- The main measuring device is equipped with a 0.87-inch small LCD screen, which displays the three-axis acceleration and battery power percentage in real time;
- The speed measurement is completed in the car, and the numerical changes at different sampling points can be analyzed through the curve chart, and the vibration can be automatically monitored, with a resonance frequency of 2.4 KHz and a bandwidth of (3DB) 500 Hz;

Technical parameters:

Balance coefficient measuring precision:	±0.02
Power measuring precision:	Greater than ±1%
Frequency range:	0.5Hz~500Hz
Variable-frequency power precision:	≤ ± 1.5Hz
Power range of elevator tractor:	1 kW-30kW
Printing method:	Bluetooth printer
Number of axles of speed measuring sensor:	Three axis directions (X, Y, Z)

Elevator no-load 125% braking performance detector ET-125



电梯无载荷 125%制动性能检测仪 ET-125Elevator no-load 125% braking performance detector ET-125

Product functions:

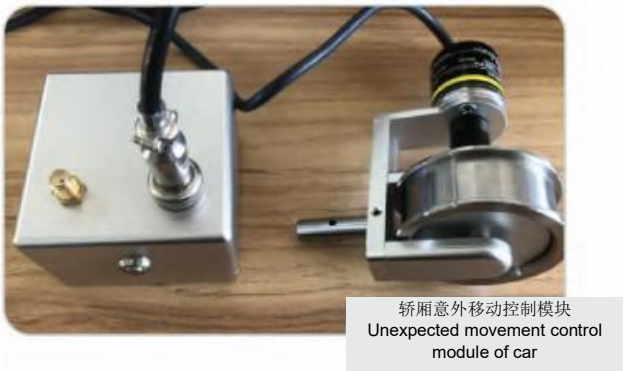
It is a kind of special equipment applicable to the evaluation and detection of the safety performance of 125% rated load braking test of machine-room elevator and machine-room-less elevator in a no-load mode on site. The main measuring device is connected with the control terminal through Bluetooth, and the measuring result can be triggered automatically. The measurement and analysis of the following parameters can be completed through one measurement: operating distance, operating time, braking distance, braking time, maximum speed, starting acceleration, braking deceleration, X-axis vibration peak, X-axis A95 vibration peak, Y-axis A95 vibration peak, Z-axis vibration peak, Z-axis A95 vibration peak, A95 acceleration, A95 deceleration and maximum jerk.

Performance characteristics:

- The main measuring device is configured with 3D-MEMS triaxial acceleration sensor, which can realize wireless communication with the handheld terminal.
- The main measuring device is equipped with a small LCD which can read the triaxial acceleration value in real time;
- The device is equipped with special data analysis and processing software, which supports data filtering, curve display and analysis, and it is intuitive and convenient, ensuring easy analysis and operation.
- The host shell is made of light aluminum alloy, which is light and solid;
- The whole measurement process is completed in the car, and the software can automatically determine the stop based on the results, thus the safe and efficient detection;
- Mass data storage function with storage space of 64GB;
- Chinese characters, numbers and special characters can be input at will, and the setting can be completed conveniently and flexibly;
- Automatic triggering is supported, and triggering operation will not cause related errors.

Technical parameters:

Number of axles:	Three axles
Measurement mode:	Automatic measurement
Measuring precision of acceleration:	≤ 1%
Measuring range of acceleration:	± 2g
Speed display resolution:	≤ 0.001m/s
Speed measuring precision:	≤ 1%
Non-linearity of acceleration sensor:	0.5% full range linear calibration
Data throughput frequency of acceleration sensor:	200hz
Braking distance resolution:	0.001m



电梯意外移动检测仪 ET-YWC
Elevator unexpected movement detector ET-YWC

Product functions:

It is a special device suitable for detecting the unexpected movement of car. The main measuring device is connected with the control terminal via Bluetooth. The software interface of this instrument is friendly, simple and easy to use, with high measuring precision and convenient and safe operation. Operators can use it without mastering special measuring knowledge. It applies to the on-site detection of unexpected movement of car by quality supervision bureau and special inspection unit.

Performance characteristics:

- The sensors used in each module of this instrument are high-precision, and feature high sampling precision and fast sampling speed, effectively improving the measuring precision of the instrument;
- It can detect the moving distance under the specified speed for the test of car unexpected movement protection device, and measure the running speed, time and distance of the elevator when UCMP acts;
- The instrument does not intervene in the elevator control system during the measurement and does not affect the normal operation of the elevator;
- Two speed-time 2D curves of traction sheave and wire rope are generated in real time, and the measurement process is more intuitive and accurate;
- Control system platform: Windows operating system. It can intelligently analyze and generate the measurement report after the completion of measurement, the measurement data has its own time attribute, and the system can print the test report on site by wireless printer;
- USB2.0 communication, no driver installation, and easy data sharing;
- External 4-segment battery power display indicator;
- The wireless transmission mode is adopted for the main body of the host and the main data collector, and no additional signal transmission antenna is required;
- The main engine shell is made of light aluminum alloy, which is light and solid;

Technical parameters:

Speed measuring range:	0~10m/s (0~6000r/min)	Displacement measuring precision:	≤ ± 0.1mm
Speed measuring precision:	≤1%	Speed display resolution:	0.001m/s
		Number of saved test groups:	No less than 99,999 groups

电梯安全回路检测仪 ET-ANT
Elevator safety loop detector ET-ANT

Product functions:

It is used to detect the design defects in the old elevator control circuit, and the short-circuited and failed key circuits under certain conditions (excessive humidity in moisture regain weather, failure of insulation to ground, illegal wiring in maintenance, etc.).

Performance characteristics:

- Test the effectiveness of elevator door lock and other safety device circuits, adaptively judge the type of AC/DC circuit, and automatically measure the function of intelligent detection system;
- Conduct non-destructive and rapid simulation of key circuit grounding fault under real-time embedded operating system;
- Conduct high-speed multi-channel precision data acquisition and multi-channel precision electronic load distribution;
- Automatically judge and adapt to the types of AC and DC circuits, and automatically measure various parameters in the input circuit;
- The wireless transmission mode is adopted for the main body of the host and the main data collector, and no additional signal transmission antenna is required;
- Control system platform: windows operating system, which can intelligently analyze and generate measurement reports after the measurement is completed;
- and can print the test report on site by wireless printer;
- Magnetic adsorption connection is adopted, which is convenient to install and high in efficiency;
- The host shell is made of light aluminum alloy, which is light and solid;
- External 4-segment battery power display indicator;

Technical parameters:

Operating voltage peak:	≤ 400V
Fault action time:	1 - 300 ms
Operating environment:	0-40°C、≤ 80% RH

电梯极限开关检测仪 ET-KG
Elevator limit switch detector ET-KG

Product functions:

This instrument is mainly used for detecting the limit switch of elevator, monitoring the on signal and off signal generated after the car triggers the limit switch of elevator, and judging whether the limit switch is in working state. It can not only improve the inspection efficiency, but also ensure the on-site safety of inspectors.

Performance characteristics:

- Electromagnetic induction protection switch is adopted for this instrument, and the high-power spread-spectrum wireless communication function between switches simulates the on-off detection protection of limit switches.
- The real-time reflection distance is recorded and set in the elevator operation limit; the device is equipped with special data analysis and processing software, which supports data filtering, curve display and analysis, and it is intuitive and convenient, ensuring easy analysis and operation;
- The whole measurement process is completed in the machine room, ensuring safe and efficient detection;
- The electromagnetic induction protection switch is adopted to effectively prevent the elevator from hitting the ceiling, falling to the bottom and shearing passengers when the elevator safety loop is short circuited and the elevator limit auxiliary switch fails.
- It is configured with system setting function so as to mark calibration and detection before detection, view the detection and verification of the curve displacement change in real time, and configure port setting and configuration information. The initial function is set before the measurement starts, and this facilitates the inspection of on-site use, link and signal interference of the instrument;
- The instrument monitors the numerical changes at low-pass sampling points and automatically monitors the on and off of switch;
- Low-power processor is adopted to minimize power consumption, with reverse connection and load short-circuit protection and electrical detection line.

Technical parameters:

Speed measuring range:	0-5m/s	Update frequency:	10 times/s
Range of distance:	200m	Short circuit protection:	Automatic recovery type
Precision:	0.1mm	Resolution:	0.1mm
Non-linearity:	0.1% F.S	Temperature range:	-25 - +80C°

电梯超载测试仪 ET-CZ
Elevator overload tester ET-CZ

Product functions:

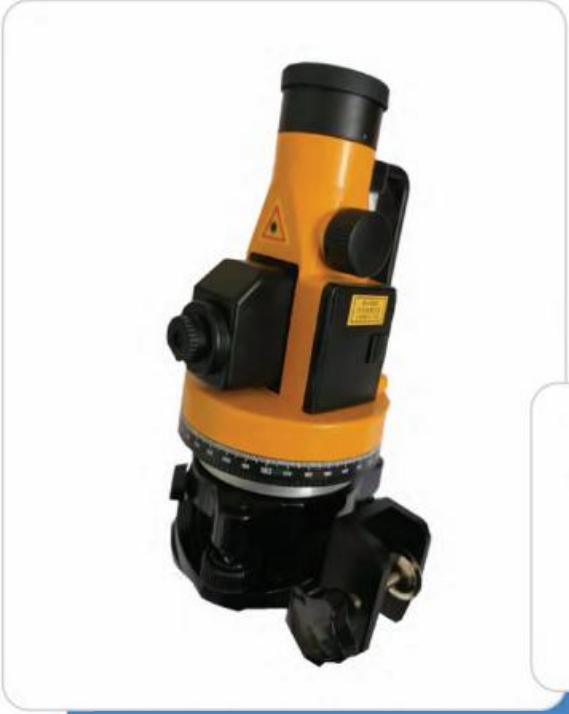
It is used to simulate the stress structure of the elevator when the car is loaded, and judge whether the overload protection device meets the standard requirements by combining the elevator's parameters and the action of the overload protection device.

Performance characteristics:

- The online detection of excessive load, spring compression and other indicators can be completed by one-time installation detection;
- Tooling is used for the detection module of excessive load and spring compression, which is easy to install;
- Recordable parameters: The handheld control terminal monitors the change of numerical parameters throughout the measurement process in real time;
- Control system platform: windows operating system, which can intelligently analyze and generate the measurement report after the measurement is completed, and can print the test report on site by wireless printer;
- The sensors used in each module of this instrument are imported and high-precision sensors, and feature high sampling precision and fast sampling speed, effectively improving the measuring precision of the instrument;
- External 4-segment battery power display indicator;
- The wireless transmission mode is adopted for the main body of the host and the main data collector, and no additional signal transmission antenna is required;
- The main engine shell is made of light aluminum alloy, which is light and solid;

Technical parameters:

Displacement resolution:	0.1mm	Force measuring resolution:	0.1N
Force measuring range:	0~5000N	Non-linearity:	0.5%F.S
Measuring precision:	± 0.5%F.S	Repeatability error:	0.5%F.S
		Safety overload:	150%F.S



电梯导轨垂直度检测仪 ET-DCB
Elevator guide rail verticality detector ET-DCB

Product functions:

This instrument can be used to test the verticality and linearity of the elevator guide rail, and can directly read out the deviation values before and after and in left and right directions of each measuring point.

Performance characteristics:

- The instrument is a high-precision laser measuring instrument integrating optics, mechanics and electricity. It is a complete elevator guide rail measurement system mainly composed of laser light source system, optical alignment system, automatic compensation system, power supply and circuit system, special guide rail fixing device, and visual alignment and digital light target;
- The laser used is a semiconductor laser with a wavelength of 635 nm, which is safe and radiation-free.

Technical parameters:

Standard deviation of upward one observation set vertical measurement:	1/45000	Laser spot diameter:	< 3mm/50m
Limit deviation of downward antipodal observation:	1/200	Coaxial error of collimation axis and vertical axis:	<5”
Angle value of long level:	20”/2mm	Coaxial error of laser axis and collimation axis:	<5”
Laser wavelength:	635nm	Automatic leveling range:	± 3°
Laser power:	10mw	Display precision:	0.01mm
Visual laser ranging:	70m	Measurement error:	± 0.03mm
Effective range of laser:	≥ 200m during the day and ＞250 m at night	Response speed:	≤ 3m/s

电梯联动试验分析仪 ET-JC
Elevator linkage test analyzer ET-JC

Product functions:

The elevator linkage test analyzer is widely used in the tension measurement and overload protection for rigging shackle, rigging for ships with movable block and crown block, elevators, ports, docks and lifting equipment. The instrument is equipped with high precision, sealed and moisture-proof tensile stress sensor, which has the advantages of convenient installation, simple operation, high overload resistance and good stability. It is widely used in the tension measurement and overload protection for rigging shackle, rigging for ships with movable block and crown block, ports, docks and lifting equipment.

Performance characteristics:

- Operation, display, driving and speed measurement devices are designed to be handheld and integrated, and it is unnecessary to connect a handheld measuring terminal on site;
- On the test site, the data in the analyzer can be automatically exported to the PC in real time;
- Manually export the data in the analyzer and save it to the PC;
- Import the historical data saved on PC into the software and display it in the form of curve;
- Filter the imported data.

Technical parameters:

Rated load:	10KN
Non-linearity:	≦ ±0.2%F.S
Repeatability:	≦ ±0.1%F.S
Full overload:	150%F.S
Ingress protection:	IP65

Configuration:

1 set of host (including signal processing, sensors and portable terminals)

1 piece of computer data connection cable

1 piece of internal hexagonal wrench

1 piece of program disk

1 set of supporting special charger

1 copy of warranty card, certificate of conformity, instruction manual and third-party verification report, etc.

电梯开门间隙检测仪 ET-DK
Elevator door opening clearance detector ET-DK

Product functions:

It is used to detect the opening clearance and force of side opening and center-parting elevator car doors and landing doors.

Performance characteristics:

- The control terminal and data acquisition processor are of intelligent integrated software design, which can complete the control and data reception of all functions, and is more convenient for on-site real-time data analysis, viewing and on-site printing of measurement data;
- The opening force can be set arbitrarily, and the width of the opening clearance can be controlled electrically, so that the door can automatically stop when reaching the set value of the opening force, and the maximum force and the width of opening clearance in the whole process can be recorded;
- The side opening door and center-parting door can be selected on the measuring interface;
- Integrated display and measurement design is adopted, and there is no need to connect a handheld measuring terminal or a tablet computer on site, thus more convenient and faster operation;
- The measured clearance and the current force are displayed at the same time;
- Touch screen display interface, which supports the input of Chinese characters, numbers and characters at will, features simple interface and easy operation, and displays the remaining power;
- Automatically connect Bluetooth printer to implement the wireless Bluetooth printing function.

Technical parameters:

Clearance measuring precision:	± 0.1mm	Non-linearity:	≤ 0.5%F.S
Clearance measuring range:	4~50mm	Repeatability error:	≤ 0.5%F.S
Force measuring range:	0~500N	Touch-screen display:	4.3-inch
Force measuring precision:	≤ 1%F.S	Size:	220mm*130mm*60mm
Force measuring resolution:	0.1N		

Elevator landing door stress deformation detector ET-M1

电梯层门受力变形量检测仪 ET-M1
Elevator landing door stress deformation detector ET-M1

Product functions:

It is used to measure the pressure applied to all types of elevator door and the permanent deformation of after applying pressure to the landing door, and is suitable for on-site inspection of elevator landing doors by special equipment inspection and testing organizations, elevator companies and elevator maintenance units.

Performance characteristics:

- The control terminal and data acquisition processor are of intelligent integrated software design, which can complete the control and data reception of all functions, and is more convenient for on-site real-time data analysis, viewing and on-site printing of measurement data;
- After the test, the safety function of elevator door is not affected;
- The measured force and action distance are displayed at the same time, and there is no need for an external handheld terminal or notebook computer onsite;
- Get rid of the previous pendulum type operation to protect the elevator landing door;
- Integrated display and measurement design is adopted, and there is no need to connect a handheld measuring terminal or a tablet computer on site, thus more convenient and faster operation;
- The main engine shell is made of light aluminum alloy, which is light and strong; the detection sensor features accurate data acquisition;
- Touch screen display interface, which supports the input of Chinese characters, numbers and characters at will, features simple interface and easy operation, and displays the remaining power;
- Automatic generation of measurement results, high compatibility, high stability, more suitable for industrial use;

Technical parameters:

Force measuring range:	0~500N	Safety overload:	120%F.S
Force measurement precision:	≤ 1%F.S	Displacement measuring range:	0-30mm
Force measuring resolution:	0.1N	Memory:	2GB
Non-linearity:	≤ 0.5%F.S	Operating current:	100ma
Repeatability error:	≤ 0.5%F.S	Communication mode:	Wireless/Bluetooth 2.0/USB2.0

电梯盘车装置检测仪 ET-DC
Elevator turning gear detector ET-DC

Product functions:

It is used to collect the various numerical parameters when the elevator is turned manually. The hand-held touch terminal can display the numerical changes in real time and monitor the whole process. It meets the requirements of relevant national inspection regulations. The instrument is of the design that makes full use of human factor engineering, conforms to the use habits of operators, and is compact and convenient and fast in inspection.

Performance characteristics:

- Recordable parameters: The handheld control terminal monitors the change of numerical parameters throughout the measurement process in real time;
- At the same time, it supports wired communication and Bluetooth communication;
- Operating system platform: windows operating system;
- The main engine shell is made of light aluminum alloy, which is light and strong; the detection sensor features accurate data acquisition;
- It automatically generate the measurement results after detection, has higher compatibility and higher stability, and is more suitable for industrial use;
- Storage function: Special memory chip is used to store the set parameters, test data and test results; the professional data analysis software can import the test data into the computer for save, and the test report is easy to print and archive;
- Query function: It can query the current memory status on site and call up the saved test records for viewing and printing.

Technical parameters:

Force measuring range:	0~500N
Measuring precision:	≤1%F.S
Force measuring resolution:	0.1N
Non-linearity:	≤0.5%F.S
Repeatability error:	≤0.5%F.S
Safety overload:	120%F.S

电梯曳引轮槽磨损检测仪 ET-CLC01
Elevator traction sheave groove wear detector ET-CLC01

Product functions:

It applies to measure the wear of all sheave grooves of the elevator. It meets the provision in Item 2.8(3) of TSG T7001-2009: The traction sheave groove shall not be seriously worn. If the abrasion of the sheave groove may affect the traction capacity, the traction capacity verification test shall be carried out.

Performance characteristics:

- It is equipped with high-precision and high resolution sensor, and displays displacement distance and deformation change synchronously. It is convenient to fully control the wear condition of the traction sheave groove;
- The measuring probe can be adjusted left and right by 0 - 15 mm according to the overall width of the traction sheave groove, which can measure all the positions on the groove;
- The handheld control terminal can collect measurement data, which is simple, effective and accurate.
- Measuring range: 6/8/10/12/13/16 mm of steel wire rope within elevator specifications (those of other specifications can be customized);
- Control system platform: windows operating system, which can intelligently analyze and generate measurement reports after the measurement is completed;
- The wireless printer prints the test report on site;
- Software requirements: integrated design, data analysis, report printing, data storage, etc.
- The handheld control terminal and the measuring unit implement wireless transmission without external connection, and the operation is convenient;
- Display mode: independent digital display of measuring unit, and zero setting function;
- Data transmission mode: wireless transmission, real-time printing of original data;
- Detachable rechargeable lithium battery is adopted, which is convenient for removal.

Technical parameters:

Measuring range:	0~15mm
Measuring precision:	≤ 0.02mm
Range:	6/8/10/12/13/16mm
Display mode:	Independent digital display
Built-in battery:	1.5 V button cell
Non-linearity:	≤ 0.05%F.S

Elevator traction sheave groove wear detector ET-CLC02

电梯曳引轮槽磨损检测尺 ET-CLC02
Elevator traction sheave groove wear detector ET-CLC02

Product functions:

It applies to measure the wear of all sheave grooves of the elevator. It meets the provision in Item 2.8(3) of TSG T7001-2009: The traction sheave groove shall not be seriously worn. If the abrasion of the sheave groove may affect the traction capacity, the traction capacity verification test shall be carried out.

Performance characteristics:

- The instrument can complete all operations through the handheld control terminal, and the data is accurate;
- Measurement range: 6/8/10/12/13/16 mm of steel wire rope within elevator specifications (those of other specifications can be customized);
- Display mode: independent digital display of measuring unit, and zero setting function;
- The measuring probe can be adjusted left and right by 0 - 15 mm according to the overall width of the traction sheave groove, which can measure all the positions on the groove;
- Detachable rechargeable lithium battery is adopted, which is convenient for removal; it has two charging modes: charging independently or charging the instrument directly, instead of disposable battery, which is energy conservation and emission reduction.

Technical parameters:

Measuring range:	0 - 15 mm (the length can be customized)
Measuring precision:	≤ 0.02mm
Measuring range:	6/8/10/12/13/16mm
Display mode:	Independent digital display of measuring unit, and zero setting function;
Built-in battery:	1.5 V button cell

Elevator guide rail gauge and coplanarity measuring instrument ET-DGM

电梯导轨轨距及共面性测量仪 ET-DGM
Elevator guide rail gauge and coplanarity measuring instrument ET-DGM

Product functions:

The instrument is suitable for on-site inspection and installation inspection of distance between guide rails and track coplanarity. It meets the requirements of relevant national inspection regulations.

Performance characteristics:

- The handheld control terminal monitors the change of numerical parameters throughout the measurement process in real time;
- Quickly and intelligently detect the absolute distance between guide rails;
- The mechanical structure is closely combined with the digital analysis system, and the angle value of coplanarity deviation between two sides is displayed digitally in real time;
- At the same time, it supports wired communication and Bluetooth communication;
- Operating system platform: windows10 operating system;
- The main engine shell is made of light aluminum alloy, which is light and strong; the detection sensor features accurate data acquisition.

Technical parameters:

Gauge measuring precision:	0.2mm
Coplanarity test precision:	0.2mm
Minimum angle resolution:	0.01°
Battery power display:	External 4-segment battery power prompt
Wireless transmission distance:	10m

电梯关门压力测试仪 ET-DM
Elevator closing pressure tester ET-DM

Product functions:

This instrument is used to measure the pinch force and kinetic energy of elevator doors; measure and display real-time impact force, impact energy, impact action time and maximum impact force parameters.

Performance characteristics:

- The control terminal and data acquisition processor are of integrated software design, which can complete the control and data reception of all functions, and is more convenient for on-site real-time data analysis and viewing;
- Follow-up measurement mode is adopted, which is convenient to operate;
- Integrated display and measurement design is adopted, which can automatically capture the maximum impact force;
- At the same time, it supports wired communication and Bluetooth communication modes;
- The main engine shell is made of light aluminum alloy, which is light and strong; the detection sensor features accurate data acquisition;
- The instrument is small and portable, anti-static, anti-interference, stable in performance, accurate in data acquisition, and does not need to be equipped with additional signal transmission antennas;
- Automatic shutdown time setting is available for users, and low-power design is adopted, thus reducing the charging times, saving energy and protecting the environment;
- Recordable parameters: Real-time digital display of pinch force and kinetic energy parameters during monitoring.

Technical parameters:

Force measuring range:	0~500N
Measuring precision:	≤1%F.S
Force measuring resolution:	0.1N
Non-linearity:	≤0.5%F.S
Repeatability error:	≤0.5%F.S
Safety overload:	120%F.S
Screen display:	3.5 inch TFT
Wireless transmission distance:	10m

电梯缓冲器测试仪 ET-DHC
Elevator buffer tester ET-DHC

Product functions:

It can detect the reset time and distance of various buffers. The detection module is small and portable, easy to use, and the time and displacement can be measured synchronously.

Performance characteristics:

- Recordable parameters: The handheld control terminal monitors the change of numerical parameters throughout the measurement process in real time;
- It automatically generate the measurement results after detection, has higher compatibility and higher stability, and is more suitable for industrial use;
- At the same time, it supports wired communication and Bluetooth communication;
- Operating system platform: windows operating system;
- The main engine shell is made of light aluminum alloy; the detection sensor features accurate data acquisition.

Technical parameters:

Distance measurement accuracy:	0.1mm
Time measurement accuracy:	≤ 0.5s
Measuring range:	0~ 2m
Time measuring range:	0~1000s
Measuring resolution:	0.1mm

电梯钢丝绳探伤仪（单根/多根）ET-G1 ET-G2
Elevator wire rope flaw detector (single/multiple) ET-G1 ET-G2

Product functions:

The flaw detector is suitable for nondestructive testing of elevator wire rope, including internal and external wire breakage, wear, corrosion, deformation, loose strands, jumping wire, material change and other defects.

Performance characteristics:

- The product meets the requirements of GB/T21837-2008, can detect local flaw (LF) and loss of metallic cross-sectional area (LMA), and can directly read out the two results in the report;
- The nondestructive testing of elevator wire rope can be completed accurately by using the methods of magnetic flux leakage and magnetic flux and the high-speed Ethernet transmission mode;
- The thin design is used to detect the plurality of densely arranged wire ropes one by one;
- Efficient military CPU is used, man-machine dialogue and data exchange with computer are carried out by wired and wireless network transmission;
- The test results can be directly read out according to the wire rope scrapping standards of different industries, and the diagnosis report is put forward;
- The computer directly samples data, processes and analyzes, displays, analyzes and prints waveforms, stores and files at one time;
- It has the characteristics of stable operation, reliable performance, high detection precision, strong anti-interference performance and accurate data;
- Computer is used to directly sample, process and analyze, and it has the functions of on-line acousto-optic alarm and real-time result display;
- The sensor is of circular structure, circumferential surrounding and axial magnetization, which instantly magnetizes the steel wire rope to a saturated state;
- It has the characteristics of stable operation, reliable performance, high detection precision, strong anti-interference performance and accurate data;
- The computer USB interface is used to directly sample, process and analyze, and it has the functions of on-line acousto-optic alarm and real-time result display;
- The sensor is equipped with alloy aluminum shell, and has the functions of waterproof, dustproof, shockproof and moisture-proof;
- The software adapts to Windows xp/7/10 and various computer operating systems;
- It has the functions of waterproof, dustproof, oil-resistant and moisture-proof;
- It is equipped with automatic safety protection and insurance structure;
- It has floating centering safety protection function.

Technical parameters:

Continuous defect (LMA) detection ability
Diameter range of measured steel wire rope: Φ1 - 30 mm (customization is available in case of special requirements)
Discontinuous defect (LF) detection ability
Allowable number of concentrated broken wires at a single position - equivalent root error

Uncertainty of LMA detection:	± 0.2%
Accuracy of LMA detection:	± 0.05%
Number of detected rope:	0~ 18m/s
Number of ropes detected simultaneously:	1 piece in a single row;
Accuracy of LF qualitative detection:	98%
Accuracy of broken wire qualitative detection:	100%
Accuracy of broken wire quantitative detection:	≥95%
Axial position indication detection ability:	± 0.3%

Comprehensive performance test system of elevator LIFT-01

Product functions:

This instrument is an intelligent comprehensive system special for multiple-performance test, and can be used to measure various parameters of elevators, such as electrical action speed, mechanical action speed, elevator acceleration and deceleration, braking distance distance between elevator car and shaft wall, elevator door opening clearance, wire rope tension, elevator landing door stress deformation, balance coefficient, buffer reset time, elevator door impact force, guide rail gauge, guide rail coplanarity, guide rail twist, escalator synchronization rate and so on.

电梯综合性能检测系统 LIFT-01
Comprehensive performance test system of elevator LIFT-01

Performance characteristics:

- A variety of signal transmission modes are adopted for the instrument, and different data transmission modes such as wired transmission, Bluetooth transmission, WIFI transmission and 4G network transmission can be selected according to the different environment of the measurement site. It fundamentally solves the difficulties in measurement position selection and measurement signal transmission on the measurement site. The instrument design is subject to the user's use, oriented to scientific measurement methods and based on strict measurement principles, which greatly improves the applicability of the instrument, the efficiency and accuracy of data measurement, and can meet the different needs of various users;
- The handheld control terminal monitors real-time figures and curves in the whole process, monitors the accuracy and stability of the inspection process and improves the analyzability of the measured data;
- It supports five communication modes at the same time, namely, wired communication, Bluetooth communication, WIFI transmission, 4G network communication and network communication.
- It supports remote synchronous double-end display of the instrument current status and measurement results;
- Battery power display: external 4-segment battery power prompt;
- Control system platform: windows operating system, which can intelligently analyze and generate measurement reports after the measurement is completed.

Technical parameters:

Governor measurement unit:

Speed measuring precision:	± 0.01m/s
Measuring range:	0~3.5m/s
Speed measurement mode:	Contact/non-contact type
Touch-screen display:	3.5-inch touchscreen

Measurement unit of door closing pressure:

Force measuring range:	0~500N
Measuring precision:	≤1%F.S
Force measuring resolution:	0.1N
Repeatability error:	≤0.5%F.S

Measuring unit of wire rope tension:

Force measuring precision:	≤1%F.S
Tension measuring range:	0-500N
Force measuring resolution:	0.1N
Non-linearity:	≤0.5%F.S
Repeatability error:	≤0.5%F
Balance coefficient measuring precision:	≤2%

Buffer reset detector unit:

Distance measurement accuracy:	0.1mm
Time measurement accuracy:	≤0.5s
Measuring range:	0~2m
Measuring resolution:	0.1mm

Measuring unit for guide rail gauge and coplanarity:

Gauge measuring precision:	0.2mm
Coplanarity test precision:	0.2mm
Minimum angle resolution:	0.01°
Weight of touch terminal:	0.35KG

Vibration analyzer unit:

Number of axles of sensor:	Three axis directions (X, Y, Z)
Acceleration measuring precision:	± 1mg
Acceleration measuring range:	± 2g
Deviation calibration:	< 1mg
Resolution:	< 1mg
Non-linearity:	<0.5%FS

Measuring unit of distance between elevator car and shaft wall:

Ranging distance:	0.15m~12m
Ranging resolution:	<0.5mm
Measuring precision:	±1mm
Range of scanning angle:	0-360°
Range of adjustable angle:	0 - 180° adjustable measuring angle
Angle resolution:	≤1°

Balance coefficient measurement unit:

Balance coefficient measuring precision:	±0.02
Measurement voltage:	25 ~ 500VAC
Measurement current:	5~ 100A
Frequency range:	0.5Hz- 500Hz
Speed measuring range:	0~20m/s
Speed measuring precision:	≤ ± 0.2%